

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. Examination (IT)

May 2013

IT706 Information Security, Security Policies & Assurance

Date: 15.05.2013, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) What is Kerckhoff's principle? List and explain the cryptanalysis attacks. [05]

(b) Why streams ciphers are suitable for GSM kind of application? [02]

Q - 2 (a) What is extended Euclidian algorithm. Find the multiplicative inverse of 87 in Z_{100} . [04]

(b) Design secret key in Play fair cipher for following characters: [07]

A-Z, 0-9, . (Dot). Do encryption for string "MY AGE IS 23."

(c) What is steganography? How it differs from cryptography? [03]

OR

Q - 2 (a) What is security mechanism? Explain the access control and traffic padding security mechanism. [04]

(b) Explain why Hill cipher is more secure than Play fair cipher. Find the inverse of following matrix over Z_{10} : [07]

$$\begin{pmatrix} 3 & 4 & 6 \\ 1 & 1 & 8 \\ 5 & 8 & 3 \end{pmatrix}$$

(c) What are peer entity and data origin security services? [03]

Q - 3 (a) Explain the round key generation in DES? Explain working of it. What is avalanche effect? [06]

(b) Explain how a substitution block cipher model as a permutation. Why block cipher cannot use full-size key? [04]

(c) What are diffusion and confusion? Explain how they affect the security of block cipher. [04]

OR

Q - 3 (a) Why AES is stronger cipher compare to DES? Explain key expansion in AES-128. [06]

(b) What is product cipher? Explain why DES uses S-boxes and D-boxes in encryption. [04]

(c) What are feistel cipher and non feistel cipher? What should be the property of mixer in feistel cipher? [04]

SECTION – II

- Q - 4 (a) Why encryptions need to be avoided if it is not required? [03]
(b) What are the requirements of cryptography hash function? List the applications of hash function in cryptography. [04]

- Q - 5 (a) Using suitable example explain how chosen cipher text attack is possible on RSA algorithm. [07]
(b) List and explain the functions used in SHA-512 hash algorithm. [04]

OR

- (b) What are the differences between CMAC and HMAC? Why these schemes can be used for authentication also? [04]
(c) Why man-in-the-middle attack is possible in Diffie-Hellman algorithm? [03]

Q - 6

- (a) Explain how keys are extracted at the sender side and receiver side in PGP. [07]
(b) What is SMIME? Explain the cryptographic message syntax in SMIME. [07]

OR

- (b) What are the services provided in SSL? Explain the procedure for the calculation of key material from master secret and cryptographic secrets from key material [07]
